CS528 Homework 1

Google Cloud Resource Management

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Cloud Console Operations

1. Experiment overview

This report documents the Cloud Console workflow for managing a storage bucket, a virtual machine, and a service account. The screenshots follow assignment operations 1-17 in order, from resource creation through cleanup.

Resource	Configuration shown in the screenshots
Project name	CS-528-YF
Project ID	dazzling-seat-508219-j3
Storage bucket	bu-cs528-fangyd-hw1-508219-j3
Bucket location / class	us-central1 (Iowa) / Standard
Directory	hw1-dir/
Uploaded file	hw1-test.txt; 20 B; text/plain
Virtual machine / zone	hw1-vm / us-central1-a
Service account	hw1-sa@dazzling-seat-508219-j3.iam.gserviceaccount.com
Bucket access role	Storage Object Admin

Workflow and evidence

Assignment operations	Evidence	Pages
1-6	Create the bucket and directory, upload and list a file, then delete the directory and its contents.	2-4
7-12	Create, suspend, resume, stop, list, and delete the VM.	4-7
13-16	Create the service account, grant bucket access, list accounts, and delete the homework account.	7-9
17	Delete the homework storage bucket.	10

Observed outcome

The screenshot sequence shows the uploaded object and the VM state changes. The final views show an empty VM list, deletion of hw1-sa, and removal of the homework bucket while the other displayed service accounts and buckets remain.

2. Bucket and directory creation

Operation 1. Make a storage bucket

The Console confirms creation of **bu-cs528-fangyd-hw1-508219-j3** in us-central1.

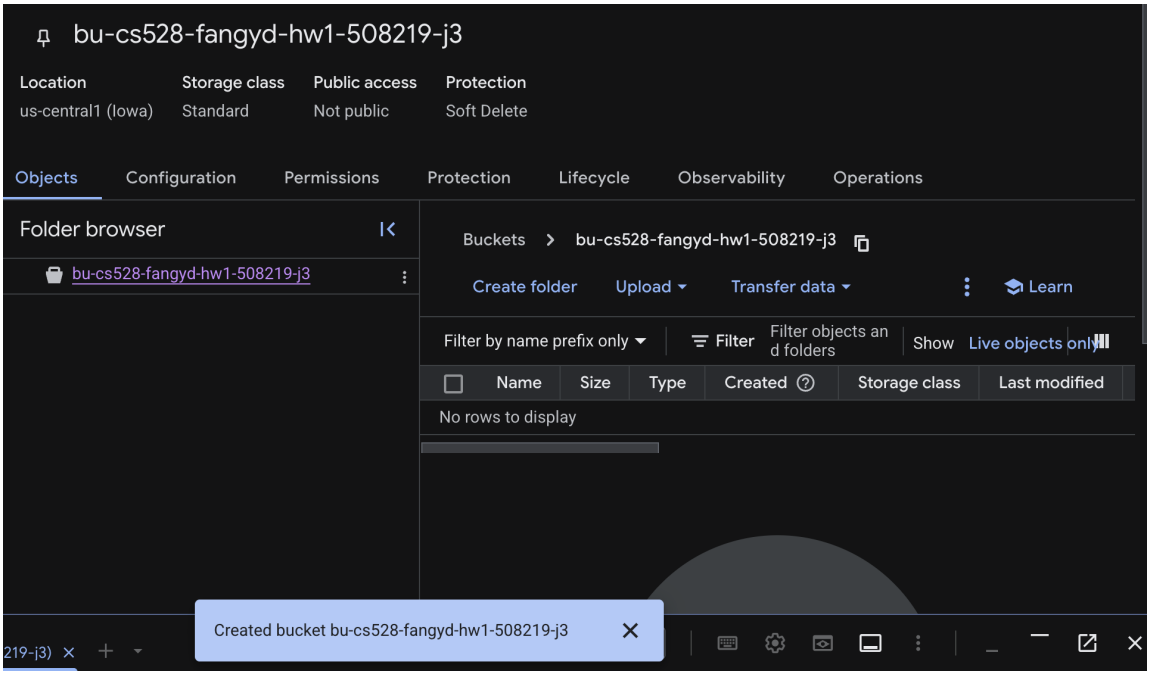


Figure 1. Created bucket with Standard storage, Not public access, and an empty object list.

Operation 2. Make a directory inside the bucket

A directory named **hw1-dir/** appears in the bucket object view and folder browser.

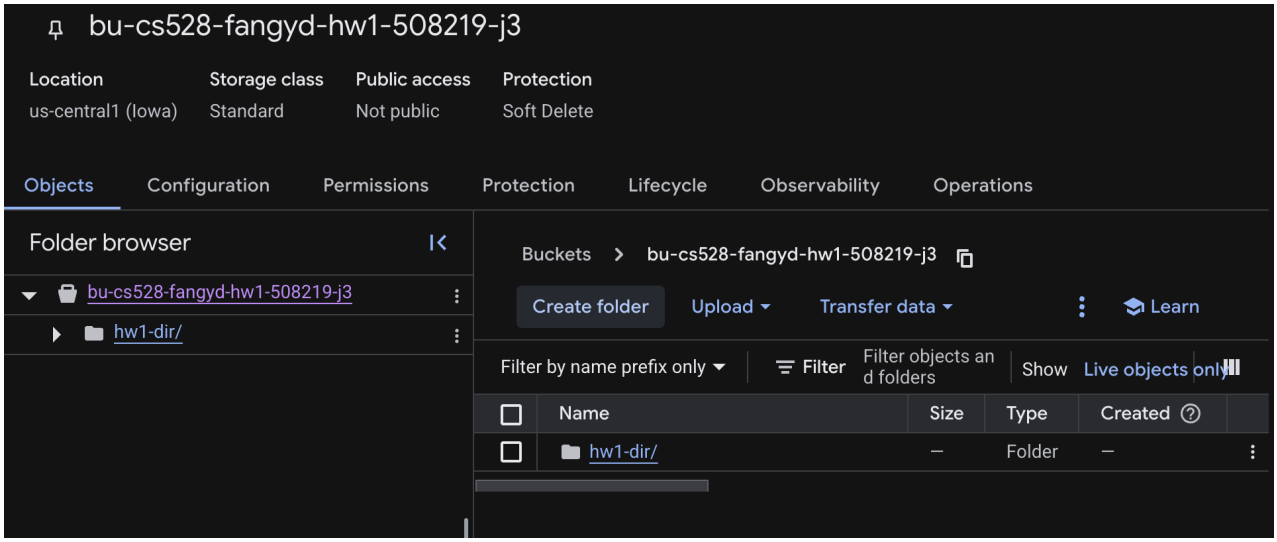


Figure 2. The newly created hw1-dir/ directory.

3. File upload and directory listing

Operation 3. List the empty directory

Opening **hw1-dir** shows no files before the upload.

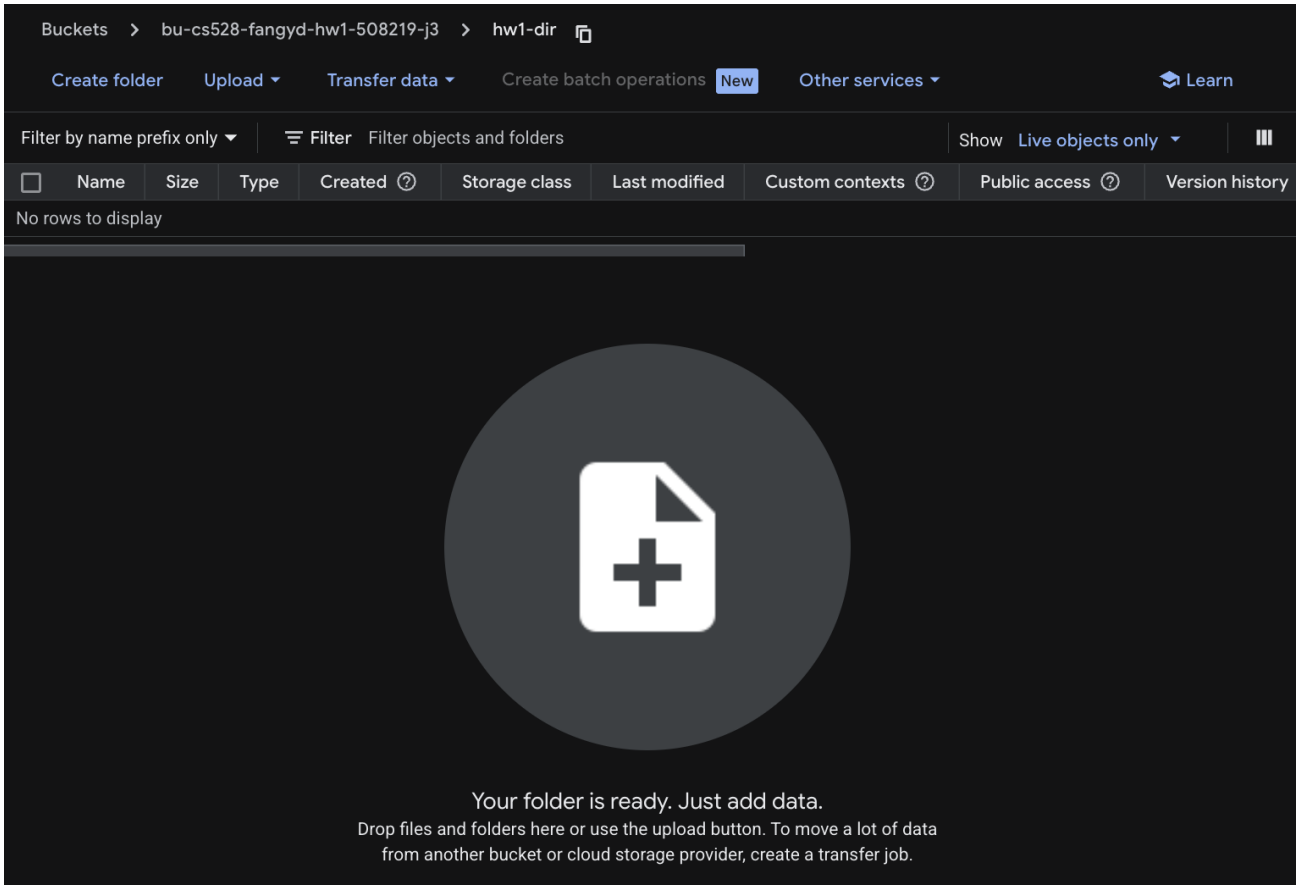


Figure 3. The directory displays "No rows to display" before a file is added.

Operations 4-5. Copy a file and list the directory

After uploading **hw1-test.txt**, the listing shows one 20-byte text/plain object.

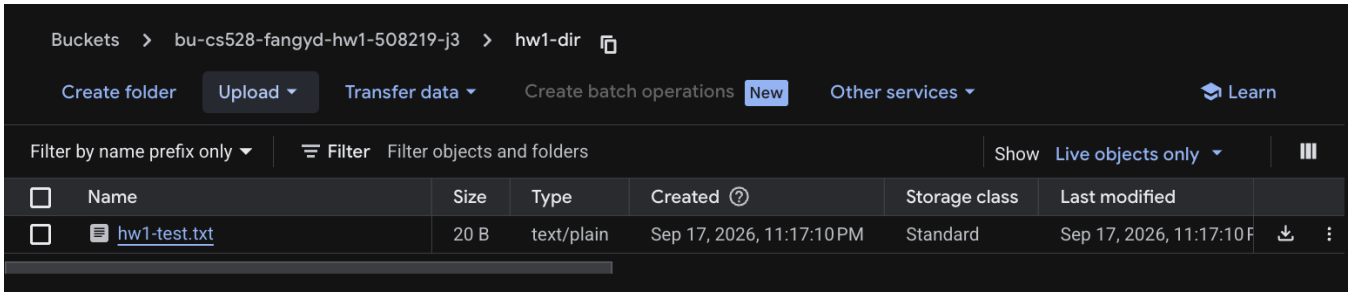


Figure 4. The uploaded file is visible inside hw1-dir/.

4. Directory cleanup and VM creation

Operation 6. Delete the directory and its file together

The directory is selected for deletion. The subsequent bucket view contains no rows.

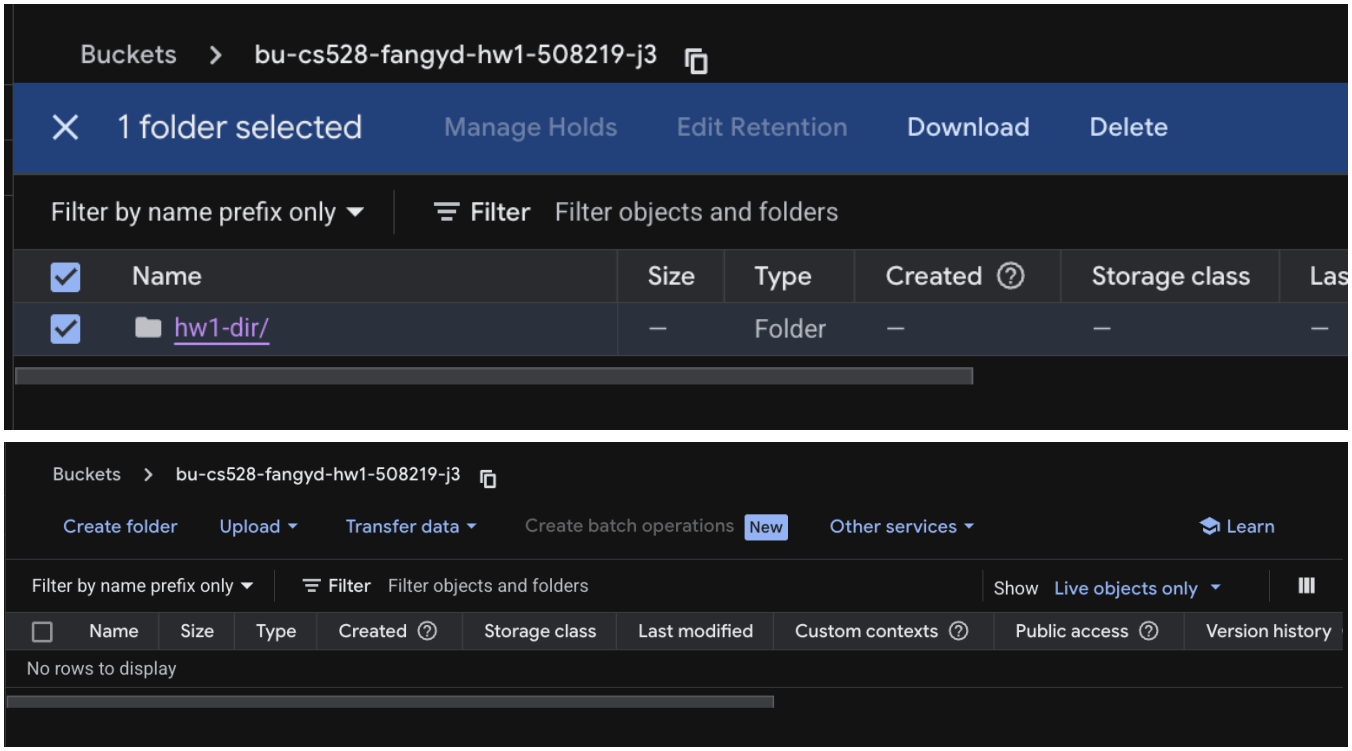


Figure 5. Folder-level deletion selection and the empty bucket view afterward.

Operation 7. Make a VM

The VM instances list shows **hw1-vm** running in **us-central1-a**.

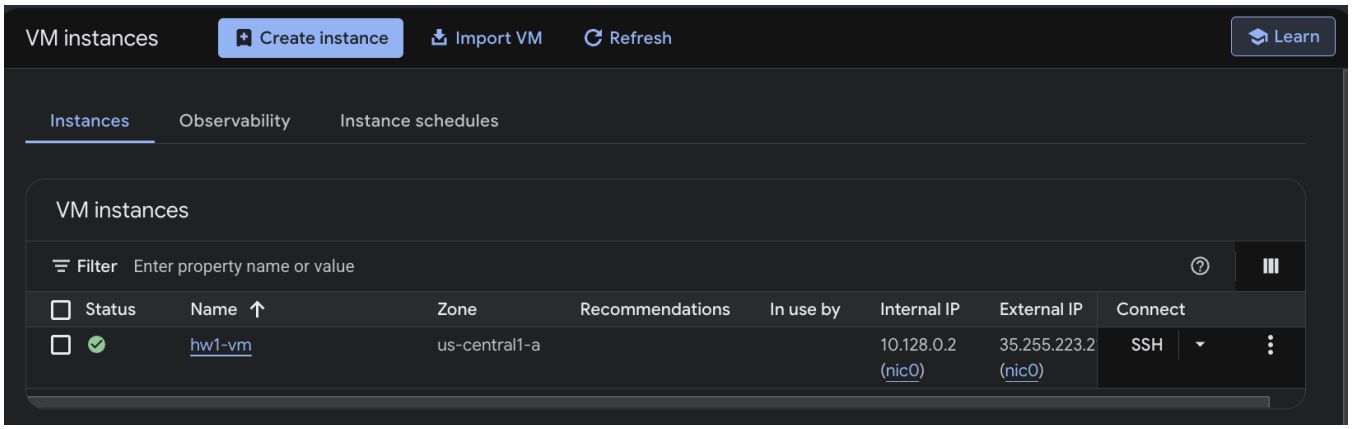


Figure 6. The created VM appears with a green running-state indicator.

5. VM suspension and resumption

Operation 8. Suspend/Pause the VM

The selected VM shows a yellow pause indicator, with **Start / Resume** available.

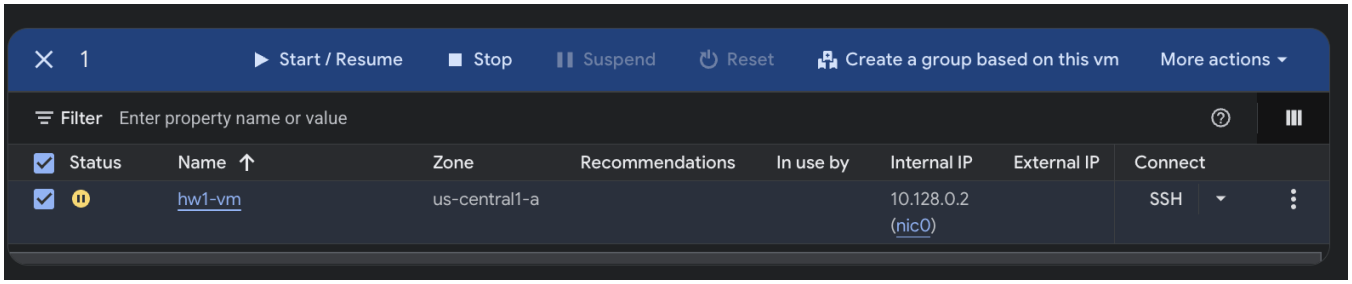


Figure 7. The VM in the suspended state.

Operation 9. Resume the VM

After resumption, the VM shows a green running indicator and the Console reports **"VM instance started"**.

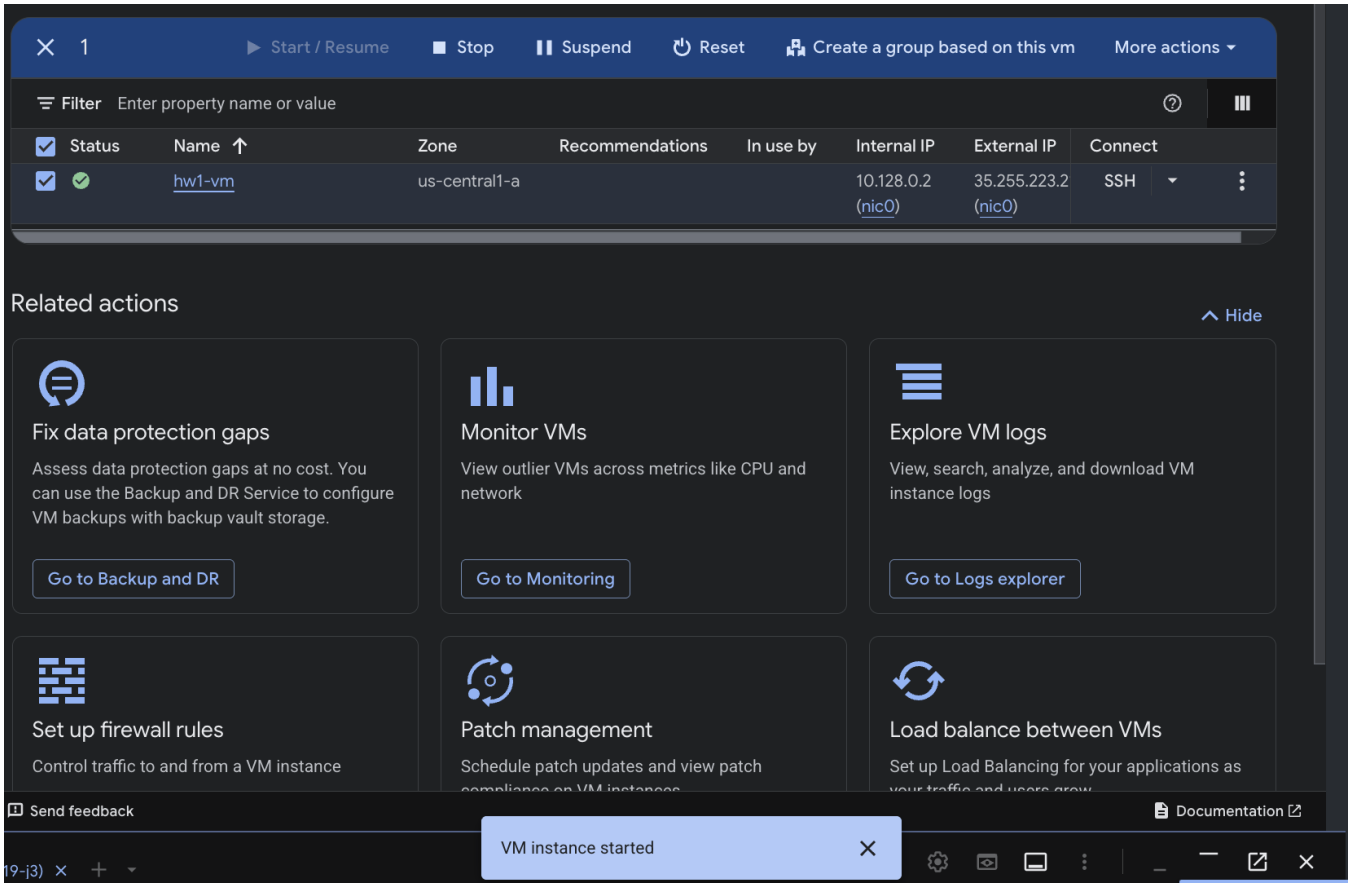


Figure 8. The resumed VM is running again.

6. VM stop and listing

Operation 10. Stop the VM

The VM is stopped; the selected row shows a gray status indicator.

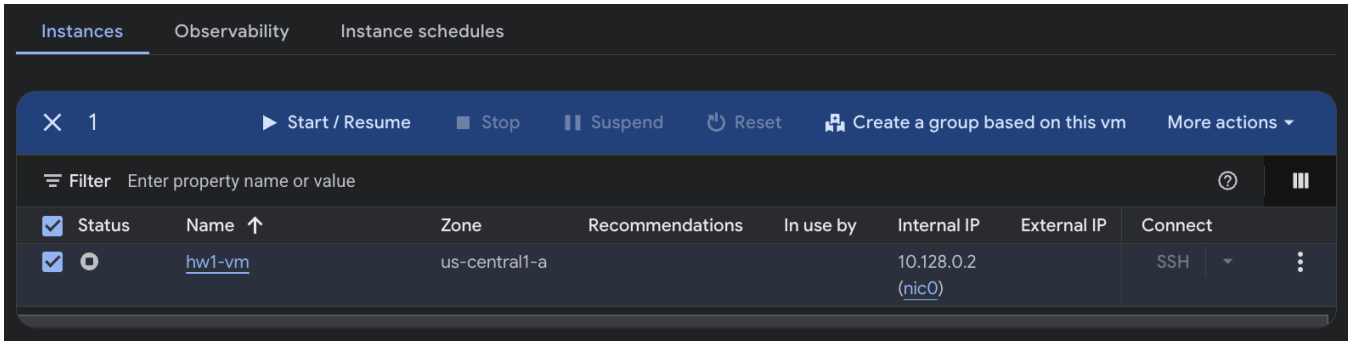


Figure 9. hw1-vm after the stop operation.

Operation 11. List all VMs

The VM instances page lists **hw1-vm** in its stopped state.

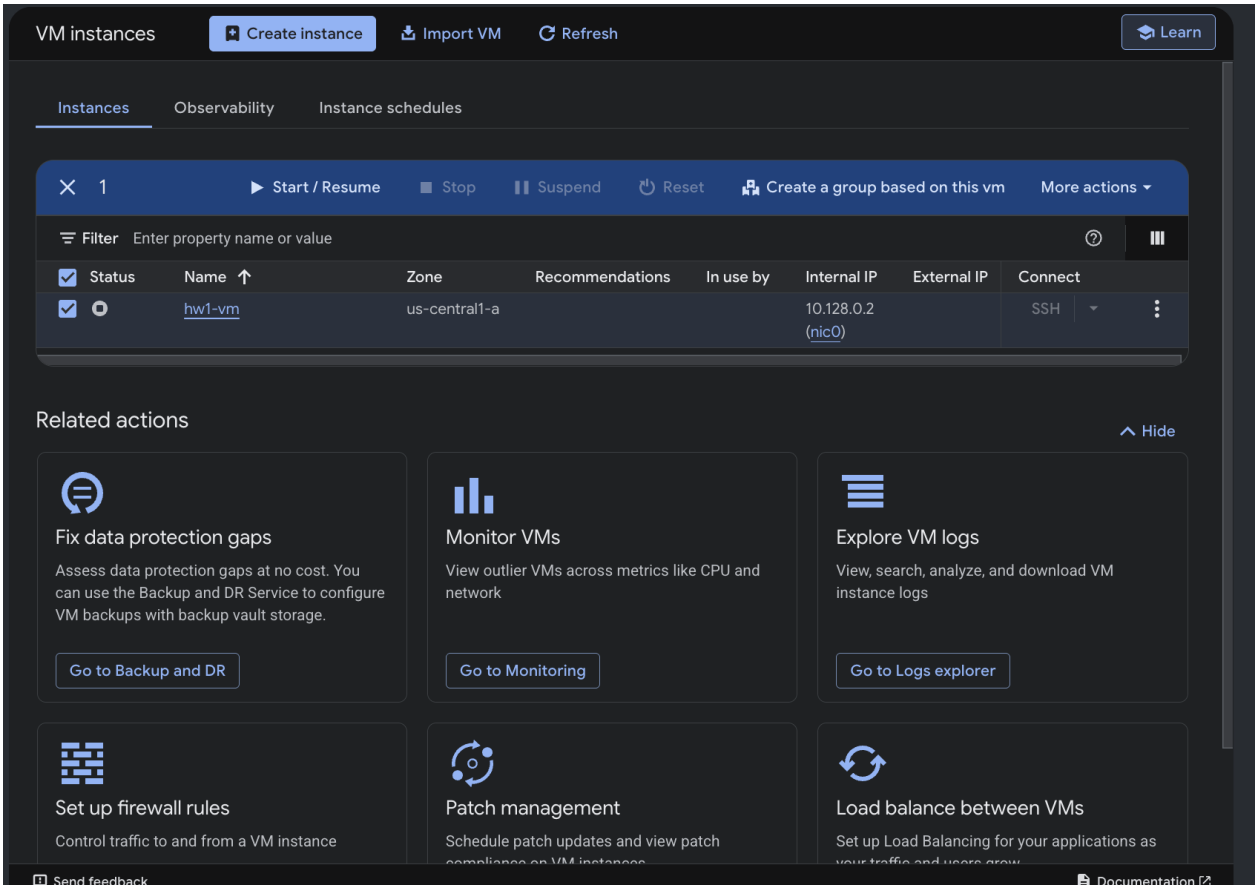


Figure 10. The VM list before the deletion step.

7. VM deletion and service account creation

Operation 12. Delete the VM

After deletion, the VM instances view contains no VM rows.

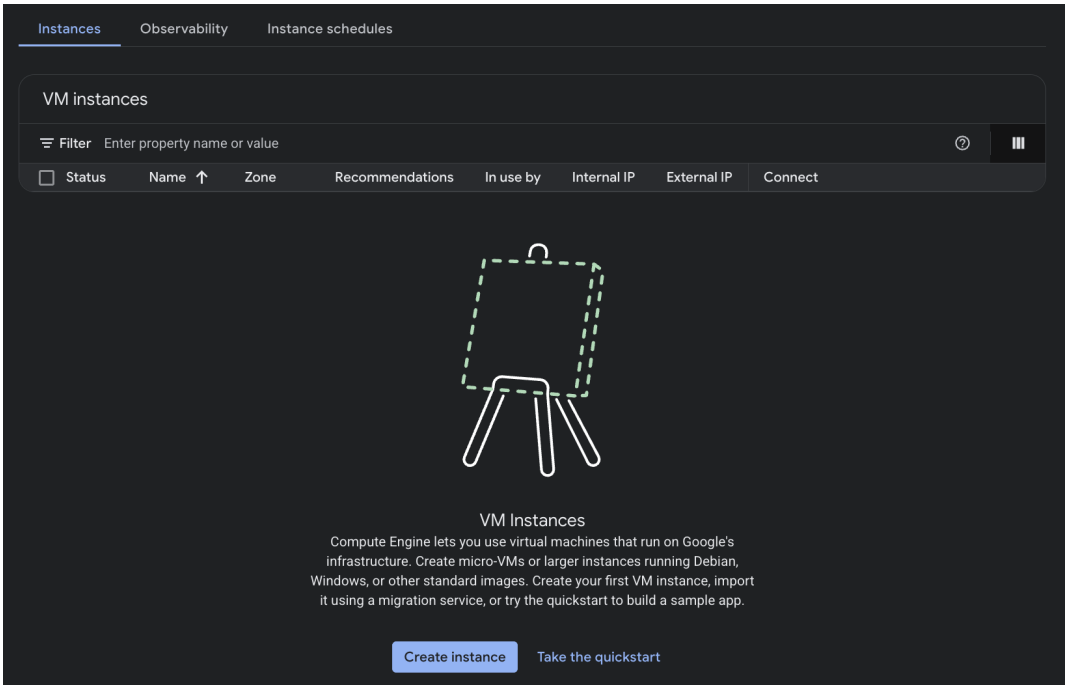


Figure 11. The empty VM instances view after deleting hw1-vm.

Operation 13. Make a service account

The service account list includes an enabled account named **hw1-sa**.

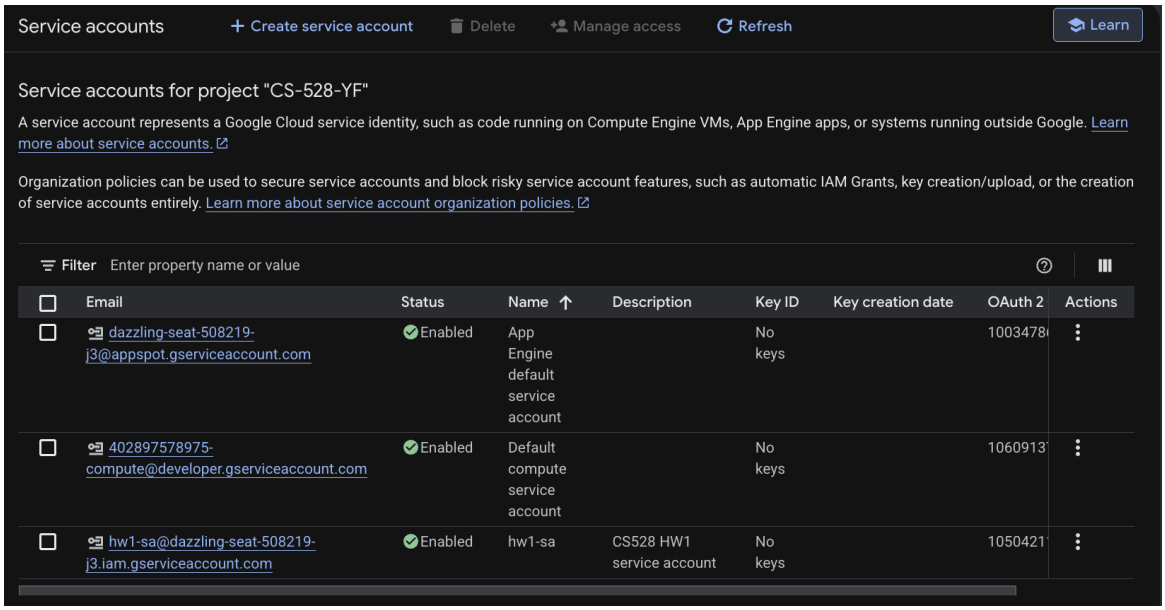


Figure 12. hw1-sa appears with the description "CS528 HW1 service account".

8. Granting bucket permissions

Operation 14. Give the service account bucket access

In the homework bucket permissions view, **hw1-sa** is selected as the principal and assigned the **Storage Object Admin** role. The saved permissions table shows the account and this role.

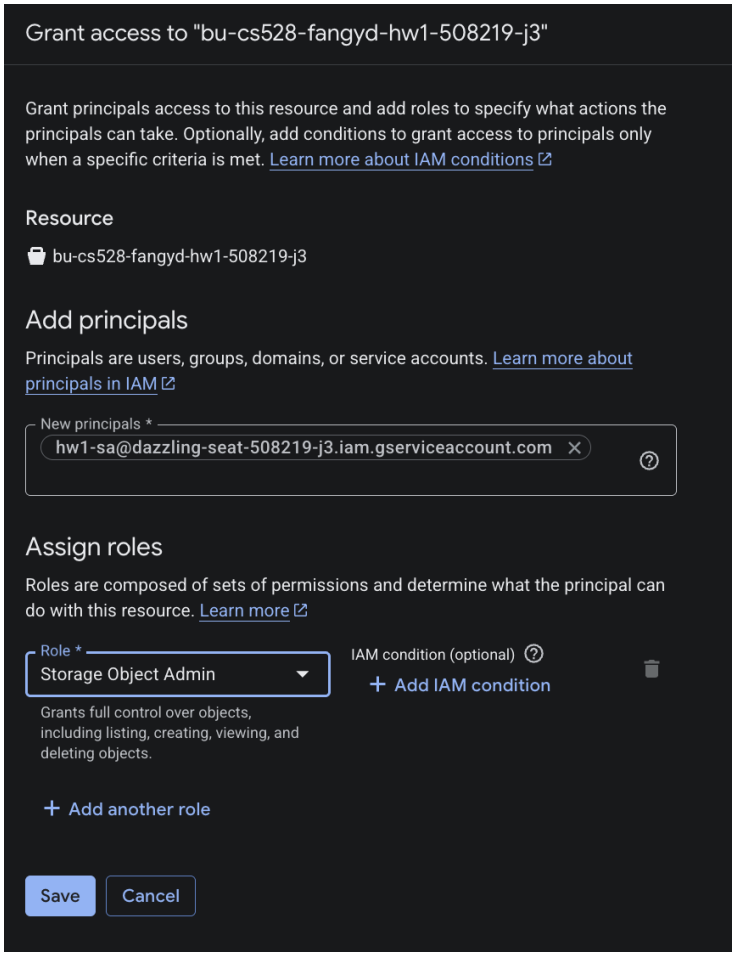


Figure 13. Bucket access configuration for the homework service account.

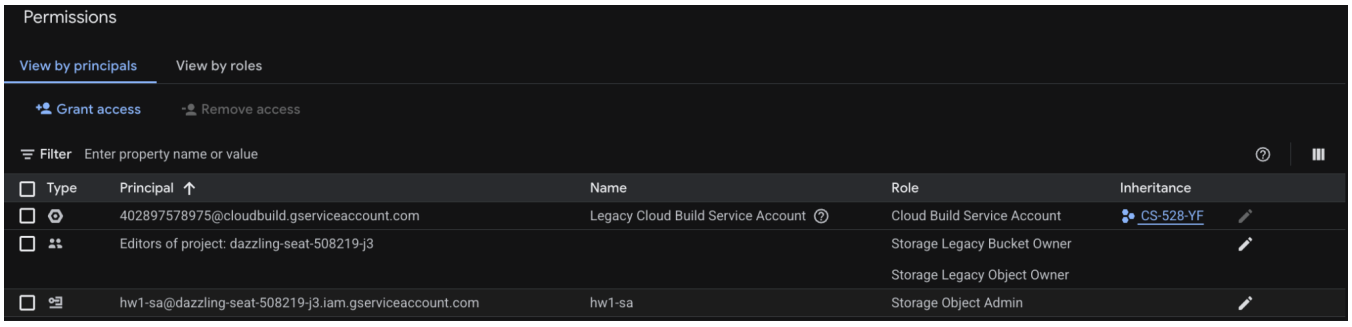


Figure 14. The bucket permissions table includes hw1-sa with Storage Object Admin.

9. Service account listing and deletion

Operation 15. List all service accounts

The list shows **hw1-sa** alongside the App Engine and Compute Engine default accounts.

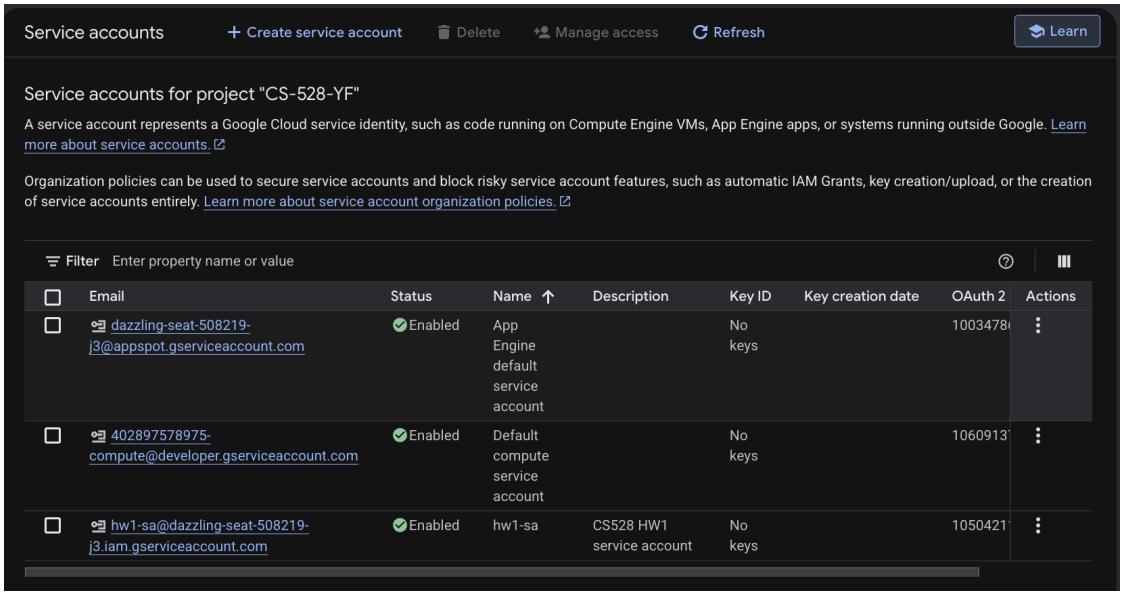


Figure 15. Service accounts before deleting the homework account.

Operation 16. Delete the service account

The Console confirms deletion of **hw1-sa**; the two default accounts remain in the list.

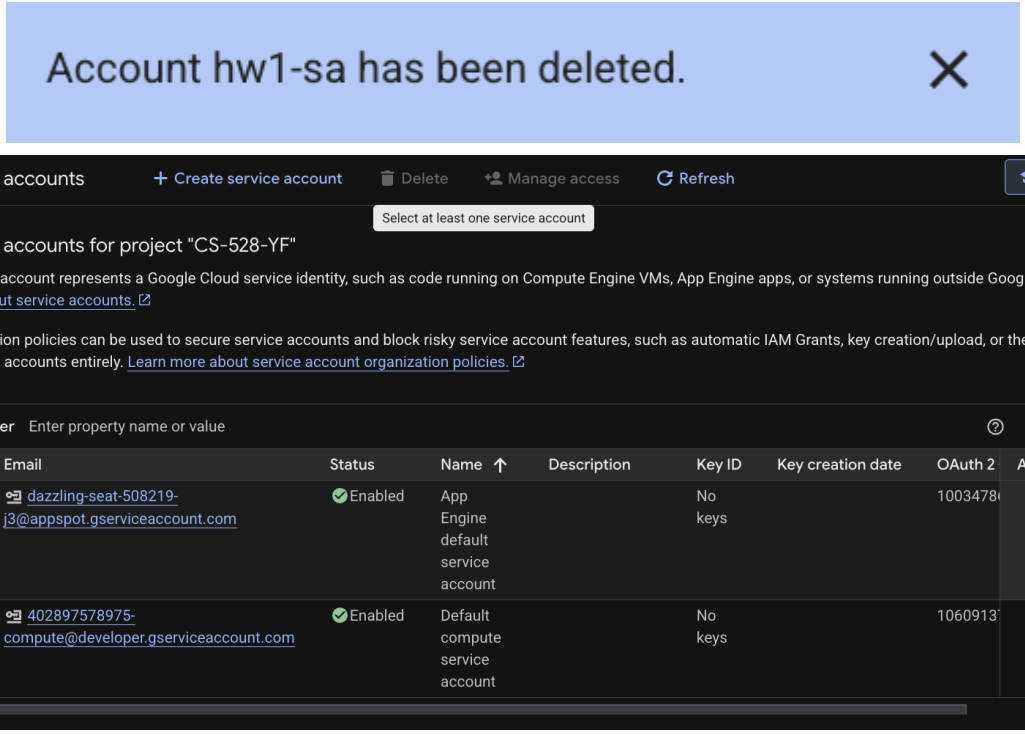


Figure 16. Deletion confirmation and the updated service account list.

10. Bucket deletion and final observations

Operation 17. Delete the storage bucket

The homework bucket is removed. The Console displays **"Deleted 1 bucket"**, and the resulting list contains the two other displayed buckets.

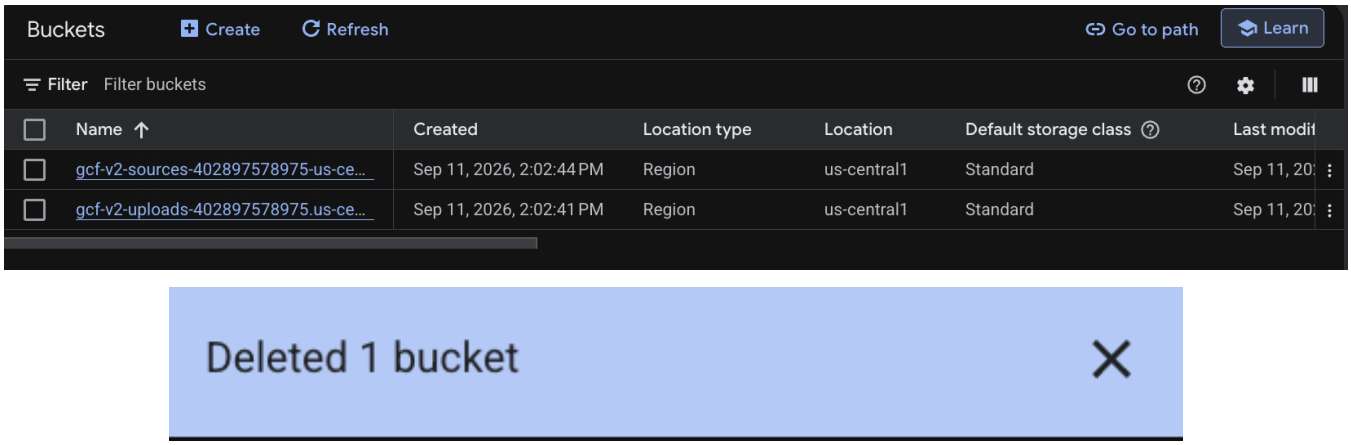


Figure 17. Bucket listing and deletion confirmation after the cleanup step.

Final observed resource state

Resource	Evidence after cleanup
hw1-dir/ and hw1-test.txt	The bucket object list is empty after directory deletion (Figure 5).
hw1-vm	The VM instances view contains no VM rows (Figure 11).
hw1-sa	Deletion is confirmed and the account is absent from the updated list (Figure 16).
Homework bucket	Deletion is confirmed and the bucket is absent from the resulting list (Figure 17).

Conclusion

The Cloud Console sequence demonstrates storage creation, file upload and listing, VM lifecycle operations, and service account access management. The final screenshots document cleanup of the homework directory, VM, service account, and bucket.